Sabina Sagynbayeva

Curriculum Vitae

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RESEARCH INTERESTS

I am an astrophysicist who primarily works on the past, present, and future of exoplanets – I study planetary dynamics and planet formation to understand the architecture of exoplanetary systems. In parallel, I also work on stars that host those exoplanets and stellar characterization by studying their surface activity using the time-series data from *Kepler* and *TESS* telescopes. I am broadly interested in general analytical techniques, astro-statistics, and numerical simulations.

EDUCATION

2021 - PRESENT **PhD:** Physics (Astrophysics)
Stony Brook University, Stony Brook, NY

2021 - 2023 **MA:** Physics (Astrophysics)
Stony Brook University, Stony Brook, NY

2016 – 2020 **Bachelor of Science:** Physics
Minor: Mathematics, Literature
Nazarbayev University, Astana, Kazakhstan

RESEARCH POSITIONS

2021 - PRESENT **Stony Brook University:** Graduate Student Researcher
Advisors: Phil Armitage, Will Farr

2022 - PRESENT **Flatiron Institute:** Guest Researcher

2025 **Kavli Institute for Theoretical Physics:** Graduate Research Fellow
Advisor: Lars Bildsten

2019 **University of Cambridge, UK:** Research Intern
Advisor: Roman Rafikov

2016-2021 **Nazarbayev University:** Research Assistant
Advisor: Daniele Malafarina

AWARDS, GRANTS, & FELLOWSHIPS

2024-2025
KITP Graduate Fellowship
I was selected for the opportunity for advanced physics doctoral students to spend a minimum period of 5 months at the Kavli Institute for Theoretical Physics.

2024
Peter Kahn Prize
An award for "outstanding research" for which I was nominated by my academic advisor.

2023
The Other Worlds Laboratory Exoplanet Summer Program
I was selected for the program that allows to visit UC Santa Cruz for three weeks to work on a project with an UCSC faculty.

2023 – 2024
Frontera Computational Science Fellowship
1-year fellowship for graduate students with an opportunity to compute on Frontera.

2022 – 2024
LSSTC Data Science Fellowship Program
I was selected for the program that consists of six week-long sessions on data science.

2020
Young Researchers Alliance FRIP program
The stipend awarded to students for research projects. Stipend: $1,000

2019
Yessenov Foundation Scholarship
Awarded to ten best students from Kazakhstan for a research internship in the US and European universities and laboratories. Funding: $7,500

SELECTED INVITED (16) & CONFERENCE TALKS

CIERA Observer's seminar, Northwestern University — Dec 2025

Exoplanets, Stars, and Planet Formation, Space Telescope Science Institute — Dec 2025

Carnegie EPL Astronomy Seminar, Carnegie Institution for Science — Oct 2025

Stars & Planets workshop, MIT	Aug 2025
Exoplanets group meeting, Princeton University	Jun 2025
Exoplanets group meeting, Queen Mary University, London, UK	Jun 2025
Dynamix Conference, Cambridge, UK	Jun 2025
56th DDA meeting, Atlanta, Georgia	May 2025
Planet Formation and Migration near the Inner Edge of Disks, KITP Program	Apr 2025
A&A Journal Club Talk, University of California San Diego	Apr 2025
AstroLunch seminar talk, University of California Santa Barbara	Mar 2025
PLUNCH seminar talk, University of California Santa Cruz	Mar 2025
KITP Local's Lunch, KITP	Feb 2025
Planet Formation group meeting, Flatiron Institute (CCA)	Feb 2025
MAPL Lab Group Meeting, University of California Santa Barbara	Feb 2025
Planet Formation group meeting, Flatiron Institute (CCA)	Oct 2024
Frontera Talk, Texas Advanced Computing Center	May 2024
New York Area Exoplanets Meeting (NYAEM) 2024	May 2024
Lunch Talk, Columbia University	Feb 2024
Bay Area Exoplanet Meeting 44, University of California Santa Cruz	Jul 2023
University of California Santa Barbara	Jul 2023
OWL talk, University of California Santa Cruz	Jul 2023
StanCon 2023, Washington University in St. Louis	Jun 2023
Athena++ workshop, Flatiron Institute (CCA)	May 2023
Gravitational Waves group meeting, Flatiron Institute (CCA)	Oct 2022
Seminar, University of Kansas	May 2022

TEACHING APPOINTMENTS

MAY 2023 – JUL 2023

Teaching Assistant
Course: Classical Physics Lab
Department of Physics & Astronomy, Stony Brook University

MAR 2022 – APR 2022

Group Project Leader
Women in Science and Engineering program
Stony Brook University

AUG 2021 – DEC 2021

Teaching Assistant
Course: Introduction to Planetary Sciences
Department of Physics & Astronomy, Stony Brook University

JAN 2017 – JAN 2019

Teaching Assistant
Courses: Calculus I,II,III, Linear Algebra, Ordinary Differential Equations
Department of Mathematics, Nazarbayev University

ACADEMIC LEADERSHIP AND SERVICE

MAY 2024 –

Steering Committee Member
NASA HWO Demographics and Architectures Sub-WG

OCT 2023

Executive Secretary
NASA Astrophysics Theory Program (ATP)

SEP 2021 – MAY 2023

Senator for the Department of Physics & Astronomy, Member of the Graduate DEI Committee
Graduate Student Organization, Stony Brook University

JUN 2021 – JUN 2022

Underclass person-at-large & Director of External Affairs
Physics Graduate Student Association, Department of Physics & Astronomy, Stony Brook University

SEP 2017 – MAY 2020

Physics Department Representative
Student Council of Nazarbayev University

SELECTED OUTREACH : WATCHABLE TALK

Outreach talk: The formation of gas giants MAR 2023
iTelescope Webinar Series

Outreach talk: Oceans in the Solar System AUG 2022
Astronomy on Tap, New York City

Outreach talk: How do planets form? APR 2021
Astronomy on Tap, Baton Rouge

Writer for Astrobites.org JAN 2021 – JAN 2023
A website where graduate students publish daily summaries of recent papers on astro-ph.
I also chaired the Advertising and Undergraduate Committees.

President of the Women in Physics Club APR 2018 – MAY 2020
Nazarbayev University

Organizer at the "Education for all" center OCT 2017 – SEP 2020
An organization that helps children with mental and physical disabilities.
I organized the first three inclusive musical theatre performances in Kazakhstan

COMPUTATIONAL SKILLS

PROGRAMMING / MARKUP LANGUAGES	Python, Julia, C/C++, IDL, HTML, JavaScript, Mathematica, LaTeX
STATISTICAL SKILLS	Hierarchical Bayesian Models, Gaussian Processes, MCMC sampling
HYDRO CODES	Athena++, PLUTO
N-BODY CODES	REBOUND
FRAMEWORKS / TOOLS	git, GitHub, ds9, Slurm
SUPERCOMPUTING CLUSTERS	*seawulf* at SBU, *Frontera* at the Texas Advanced Computing Center, *rusty* at Flatiron Institute

PUBLICATIONS: 1ST/2ND AUTHOR ★: IN PRESS (CLICKABLE)

15. **Sabina Sagynbayeva** et al. "Occurrence Rates Based on Chemical Abundances of K2 Planets." *In prep.*

14. **Sabina Sagynbayeva** et al. "Rotation Periods for Stars in Open Cluster NGC 6819 From Kepler IRIS Light Curves." *In review; on arXiv.*

13. **Sabina Sagynbayeva** et al. "Polka-dotted Stars II: Starsposts and Obliquities of Kepler-17 and Kepler-63." *The Astrophysical Journal* (2026).

12. **Sabina Sagynbayeva** et al. "Polka-dotted Stars: a Hierarchical Model for Mapping Stellar Surfaces Using Occultation Light Curves." *The Astrophysical Journal* (2025). ★

11. **Sabina Sagynbayeva** et al. "Requirements for Joint Orbital Characterization of Cold Giants and Habitable Worlds with Habitable Worlds Observatory." *The Astronomical Journal* (2025).

10. **Sabina Sagynbayeva** et al. "Circumplanetary Disks are Rare around Planets at Large Orbital Radii: A Parameter Survey of Flow Morphology around Giant Planets." *The Astrophysical Journal* (2025).

9. Daniele Malafarina, **Sabina Sagynbayeva** (equal contribution). "What a difference a quadrupole makes?" *General Relativity and Gravitation* (2021).

PUBLICATIONS: NTH AUTHOR ★: IN PRESS (CLICKABLE)

8. Stephen Kane et al. [including **S. Sagynbayeva**]. "Imaging Venus-like Worlds: Spectral, Polarimetric, and UV Diagnostics for the Habitable Worlds Observatory" *In review* (2025).

7. Mona El Morsy et al. [including **S. Sagynbayeva**]. "OASIS Survey Direct Imaging and Astrometric Discovery of HIP 71618 B: A Substellar Companion Suitable for the Roman Coronagraph Instrument Technology Demonstration" *The Astrophysical Journal Letters* (2025).★

6. Thayne Currie et al. [including **S. Sagynbayeva**]. "SCExAO/CHARIS and Gaia Direct Imaging and Astrometric Discovery of a Superjovian Planet 3-4 λ/D from the Accelerating Star HIP 54515" *The Astronomical Journal* (2025). ★

5. Sarah Blunt et al. [including **S. Sagynbayeva**]. "A Statistical Method for Constraining the Capability of the Habitable Worlds Observatory to Understand Ozone Onset Time in Earth Analogs." *JATIS* (2025).

4. Rachel B. Fernandes et al. [including **S. Sagynbayeva**]. "Signatures of Atmospheric Mass Loss and Planet Migration in the Time Evolution of Short-Period Transiting Exoplanets." *The Astronomical Journal* (2025). ★

3. Briley Lewis et al. [including **S. Sagynbayeva**]. "Exploring the Effects of Astrobites Lesson Plans on Undergraduate Astronomy Students." *Physical Review Physics Education Research* (2025).

2. Thayne Currie et al. [including **S. Sagynbayeva**]. "Direct Imaging and Astrometric Discovery of a Superjovian Planet Orbiting an Accelerating Star." *Science* (2023). ★

1. D. Alina et al. [including **S. Sagynbayeva**]. "Large-scale magnetic field in the Monoceros OB-1 East molecular cloud." *Astronomy & Astrophysics* (2020).